

Robot Learning: From Fundamentals to Foundation Models

Lecture 1: Introduction to Robot Learning

Oier Mees



16.02.2026

About Me

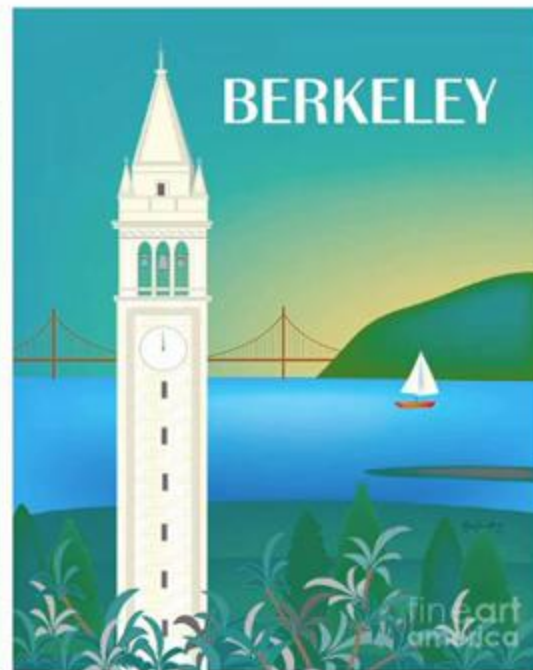
From Basque Country, Spain



PhD in Freiburg, Germany



PostDoc in Berkeley, USA



Research at Microsoft



Course Staff: TAs



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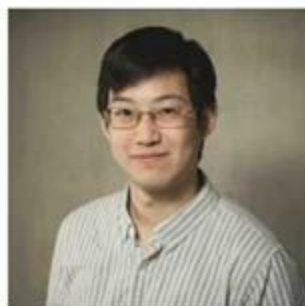
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Course Staff: TAs



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Several open TA positions:
This could be you!

The Plan for Today

- Course Goals & Logistics
- Why study Robot Learning?



Course Title: Robot Learning: From Fundamentals to Foundation Models

Semester: Spring 2026

Lecturer: [Oier Mees](#)

Teaching assistants: [Alexey Gavryushin](#), [Jonas Pai](#), [Liam Achenbach](#), [Nicola Irmiger](#),
[Tianxu An](#), [Simon Sukup](#), [Nicole Dambon](#), [Zador Pataki](#), [Carl Brandner](#)

Catalogue Link: [263-5911-001](#)

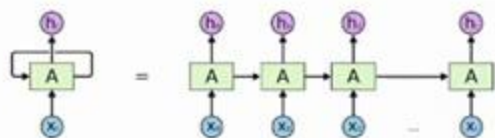
Lecture: Mon 16:15-18:00, room NO C 60.

- **Format:** Each session begins with a **lecture** on the core topic, followed by a **paper discussion** led by students. On selected weeks, we will also host short guest lectures

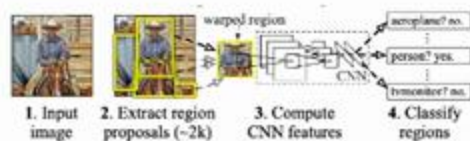
Course Goal

- **Mastery of Fundamentals:** Imitation, Reinforcement and Policy Learning
- **Practical Skills:** Hands-on simulation and real-robot policy deployment
- **Frontier Models:** Exploration of Foundation Models for robotics
- **Systems Design:** Scalable pipelines for perception, control and reasoning

Course Goal: Inspire you!



Translation: Seq2Seq...



Object Detection: R-CNN...

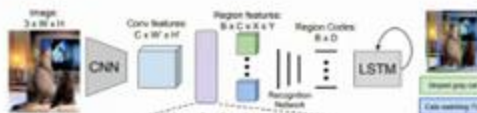
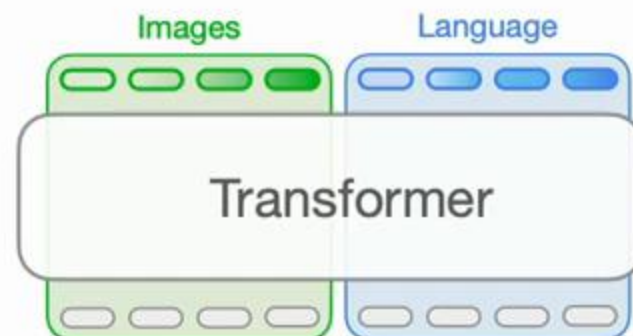
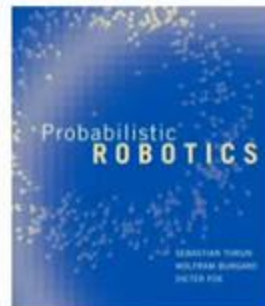


Image Captioning: DenseCap..

ChatGPT Moment



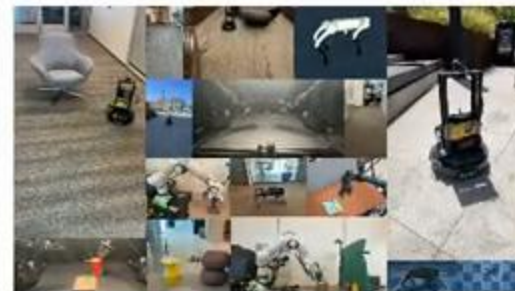
Vision-Language Models,
Generative Models...



ChatGPT for Robotics?



you



Course Structure

- Website: <https://cvg.ethz.ch/lectures/Robot-Learning>
- Mondays 16:15-17:45 :
 - Lecture 45 min + 45 min Paper Discussion
 - Guest lectures throughout course
- Thursdays 10:15-12:00
 - TA Led Practice Sessions
- Individual Homework: 4x assignments
- Group Projects on real SO-101 robots



Paper Discussion

- 3x papers per lecture, available at project course
- 15 minutes per paper, groups of 4 students
- Group A presents & defends paper, Group B criticizes it

Enter paper groups here:



<https://docs.google.com/spreadsheets/d/1YSJAvOj4riUwgBXzQlsAgiAGjVIJ3oDI4TtdlcbgJdB4/>

Lecture Tentative Schedule

Week	Monday	Paper Discussion
Week 1: Feb 16	Introduction to Robot Learning	No paper discussion.
Week 2: Feb 23	MDPs & Learning Dynamics	Simple random search provides a competitive approach to RL (Mania et al., 2018). Deep RL Doesn't Work Yet (Irpan, 2018). Curiosity-driven Exploration by Self-supervised Prediction (Pathak et al., 2017)
Week 3: Mar 02	Imitation Learning	Causal Confusion in IL (Den Haan et al., 2019). The surprising effectiveness of representation learning for visual imitation (Pari et al. 2021). Transporter Networks: Rearranging the Visual World for Robotic Manipulation (Zeng et al., 2020)
Week 4: Mar 09	Reinforcement Learning I	Evolution Strategies as a Scalable Alternative to RL (Salimans et al., 2017). Learning Synergies between Pushing and Grasping (Zeng et al., 2018). Precise and Dexterous Robotic Manipulation via Human-in-the-Loop Reinforcement Learning (Luo et al., 2024)
Week 5: Mar 16	Reinforcement Learning II	End-to-End Training of Deep Visuomotor Policies (Levine et al., 2015). Eureka: Human-Level Reward Design via Coding LLMs (Ma et al., 2023). Latent Plans for Task Agnostic Offline Reinforcement Learning (Rosete-Beas et al., 2022)

Individual Homework

- Assignments: <https://github.com/mees-robot-learning-course/ethz-course-2026>
- Submission: Gradescope

- HW1: Pytorch & Numpy Tutorial 16.02-26.02
- HW2: Robot Control & MDPs 23.02-05.03
- HW3: Imitation Learning 02.03-16.03
- HW4: Reinforcement Learning 16.03-30.03

Homework Advice

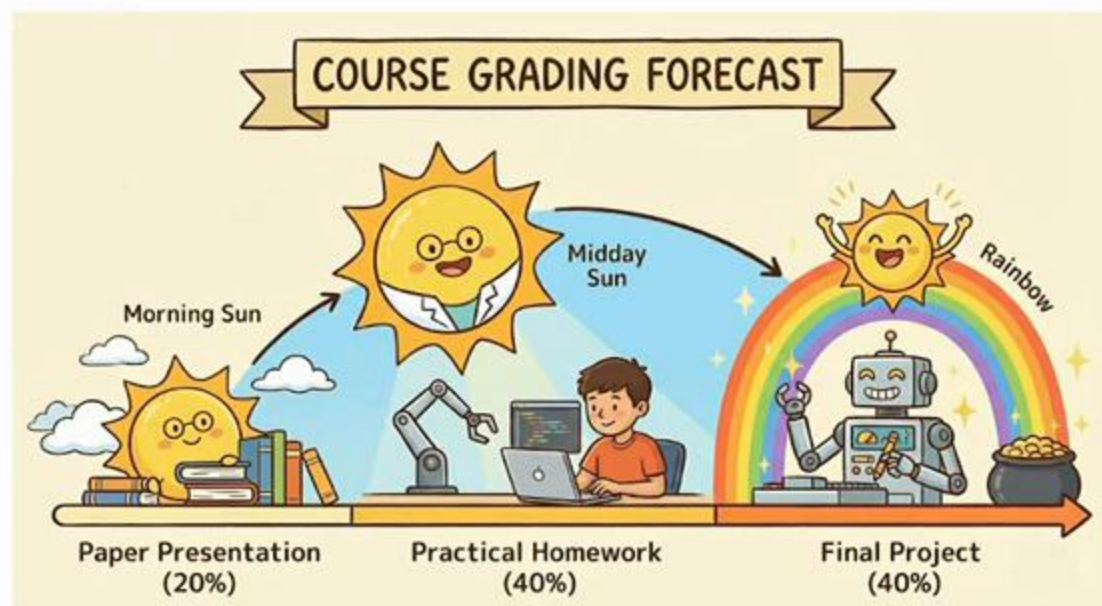
- Neural networks take some time to train!
- We try to make the homeworks fast to train



Don't start the night before the deadline working on the assignment!

Course Grading

- Paper Presentation & Discussion (Group): 20 %
- Practical Homework (Coding Assignments): 40 %
- Final Project (Group): 40 %



Feedback Welcome

- We are working hard to offer a great course
- We will probably make mistakes
- We would love to hear your feedback!



Robots in Science Fiction



Q the Automaton, 1918



Metropolis, 1927



Star Wars, 1977



The Jetsons, 1962



Transformers, 2007



Wall-E, 2008

Why Robots



Shakey the Robot



Shakey the Robot, Stanford 1966

Shakey the Robot



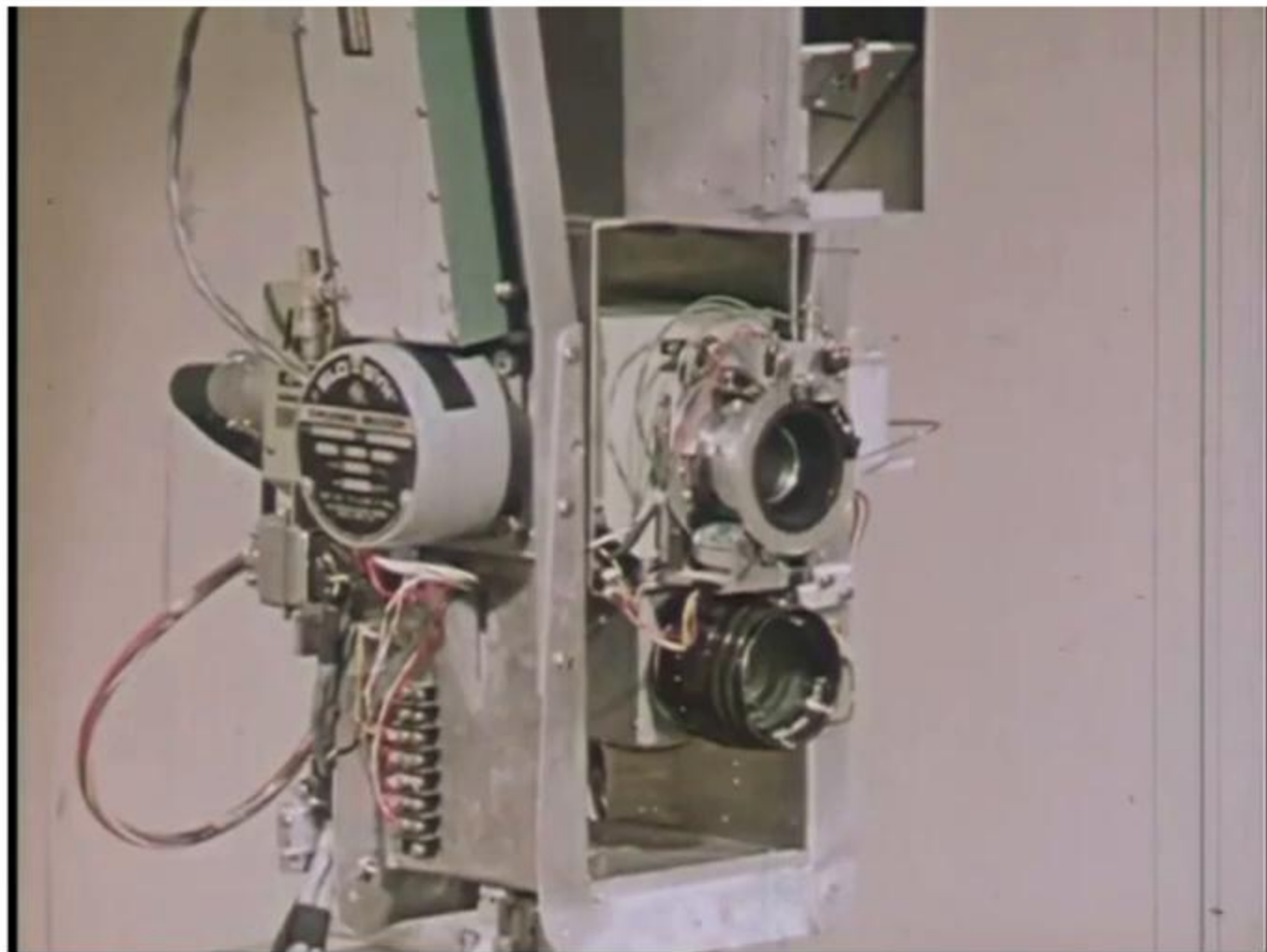
Shakey the Robot, Stanford 1966

Shakey the Robot



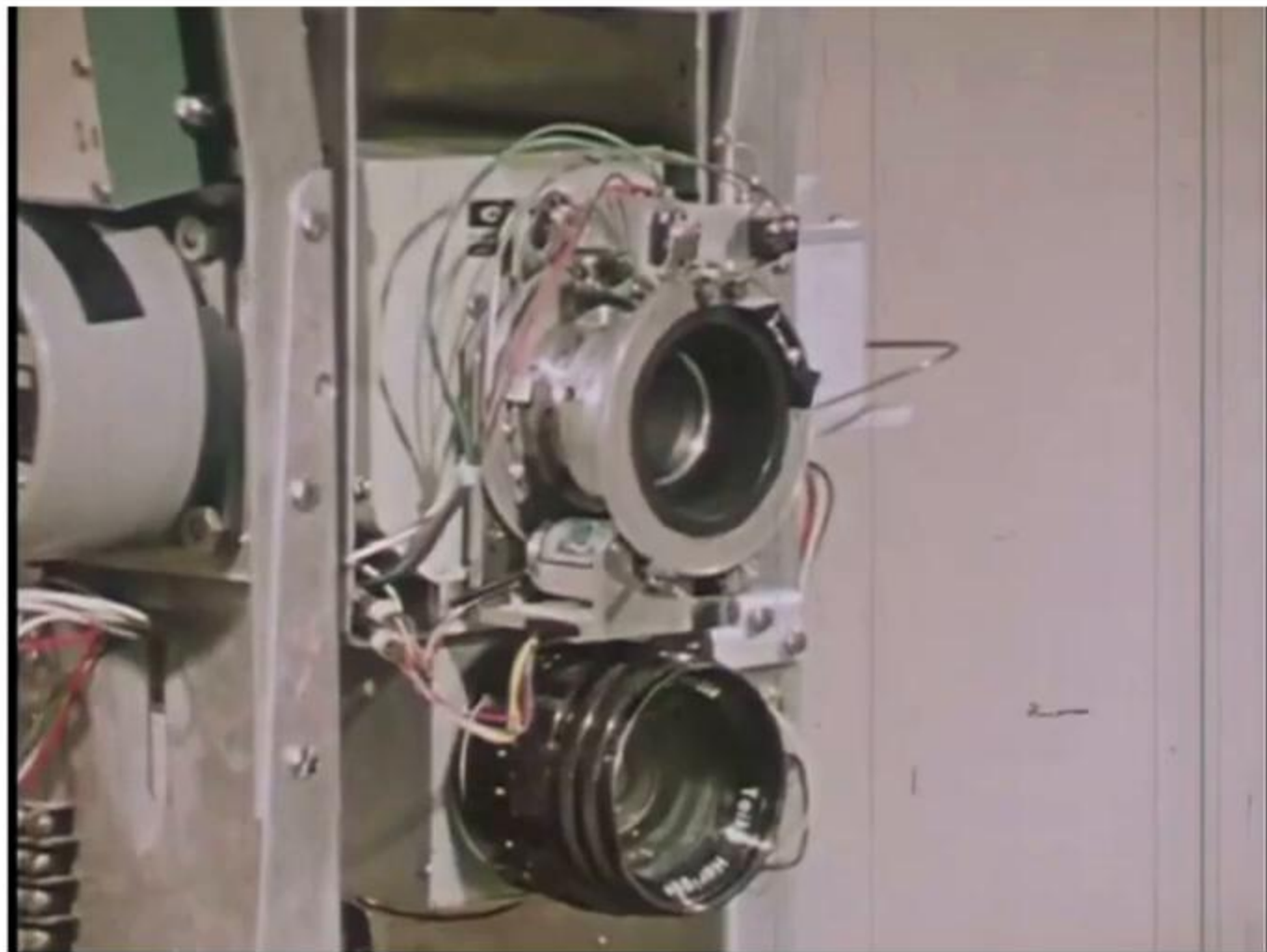
Shakey the Robot, Stanford 1966

Shakey the Robot



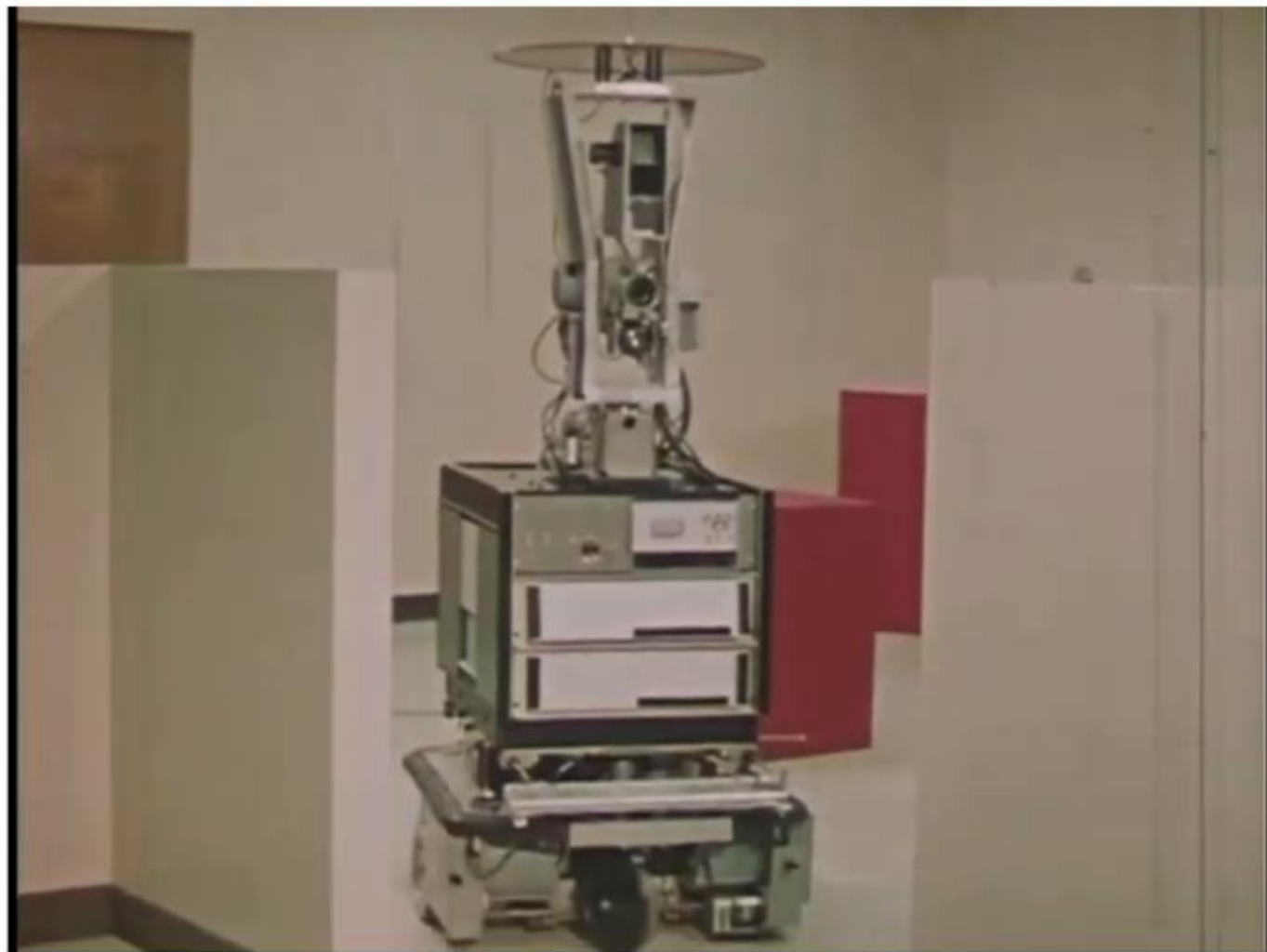
Shakey the Robot, Stanford 1966

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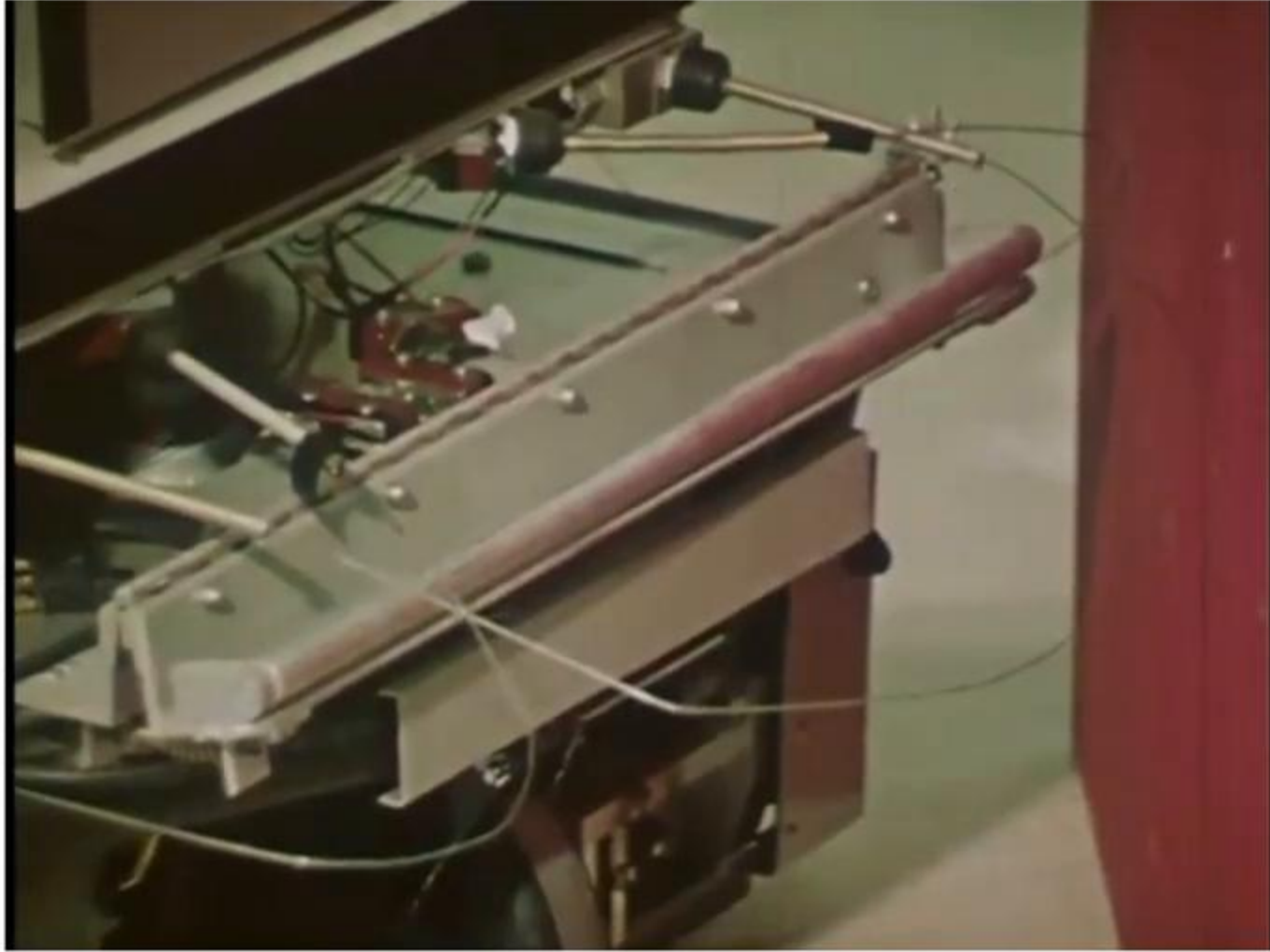
Shakey the Robot, Stanford 1966

Shakey the Robot



Shakey the Robot, Stanford 1966

Shakey the Robot



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Shakey the Robot



Shakey the Robot, Stanford 1966

Robot Engineering

- Sense-Plan-Act
- Similar to Shakey (1966)
- Structured Environments



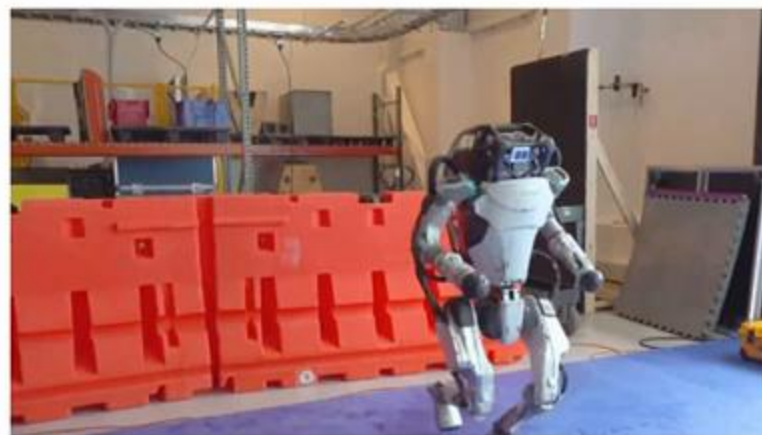
When State Estimation Fails



DARPA Robotics Challenge 2015



Boston Dynamics 2018



Boston Dynamics 2022

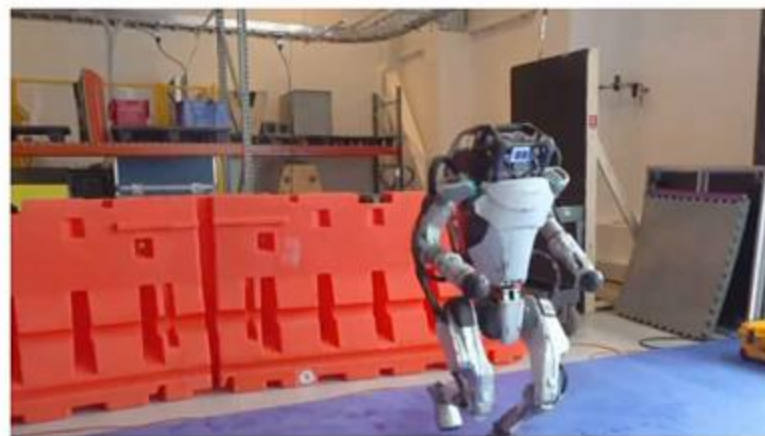
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When State Estimation Fails



DARPA Robotics Challenge 2015



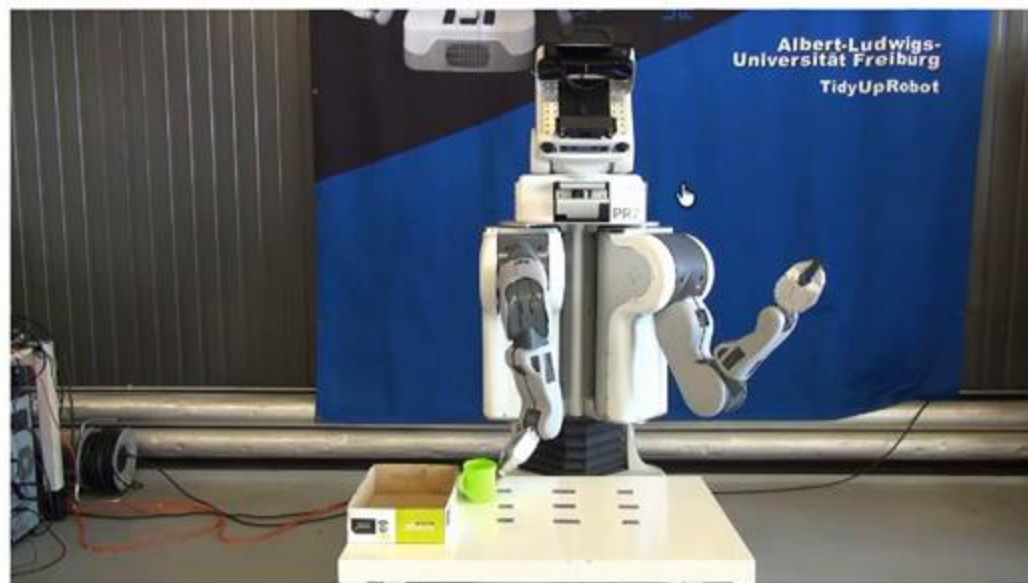
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Boston Dynamics 2022

Moravec's Paradox (1988)

- **Abstract thinking:** hard for animals, easy for AI
- **Sensorimotor skills:** easy for animals, hard for AI



Mees et al., 2019



Moravec's Paradox (1988)

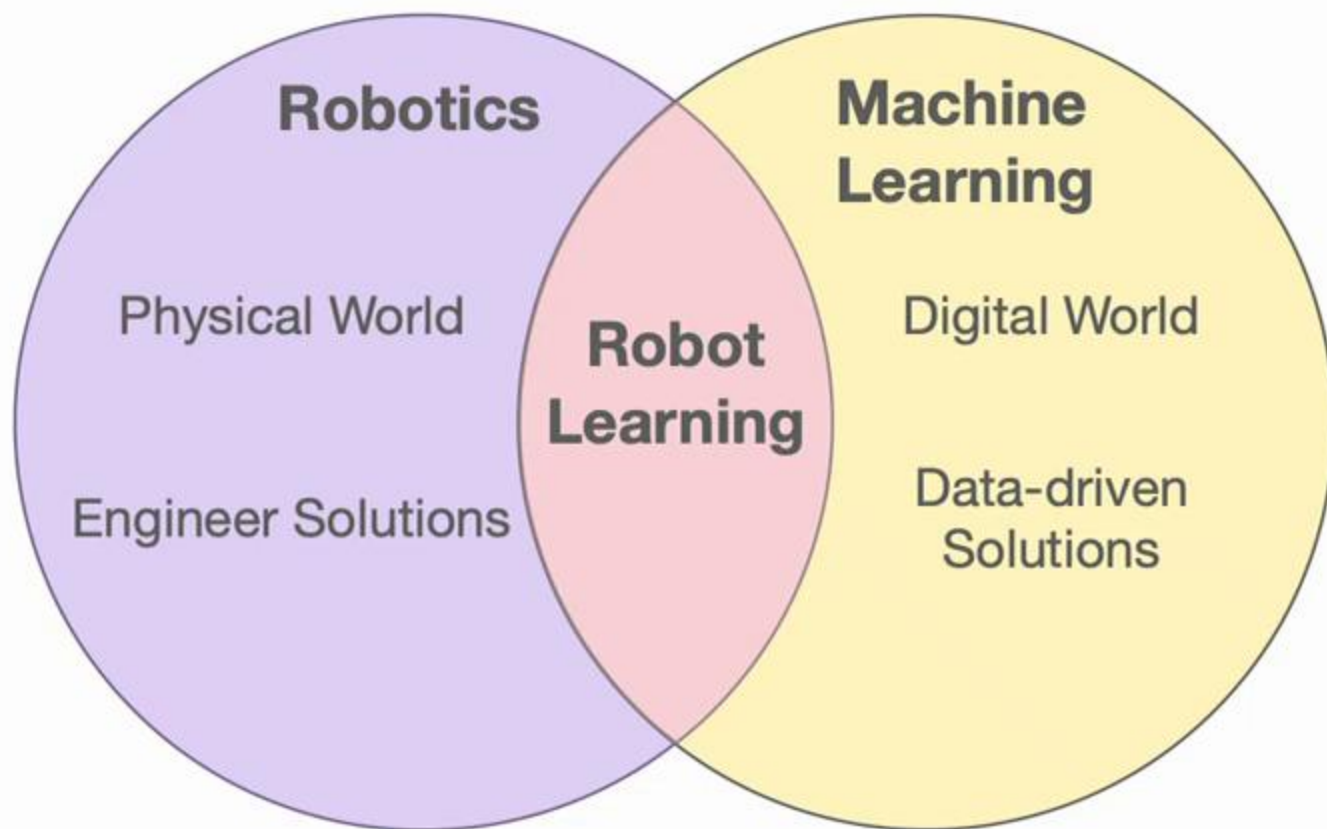
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Mees et al., 2019



Robot Learning



Robot Learning

Solving **Robotics** via **Machine Learning**



Perception
Control
...



Imitation Learning
Reinforcement Learning
Dynamics Learning
Representation Learning
...

Are These Robots?

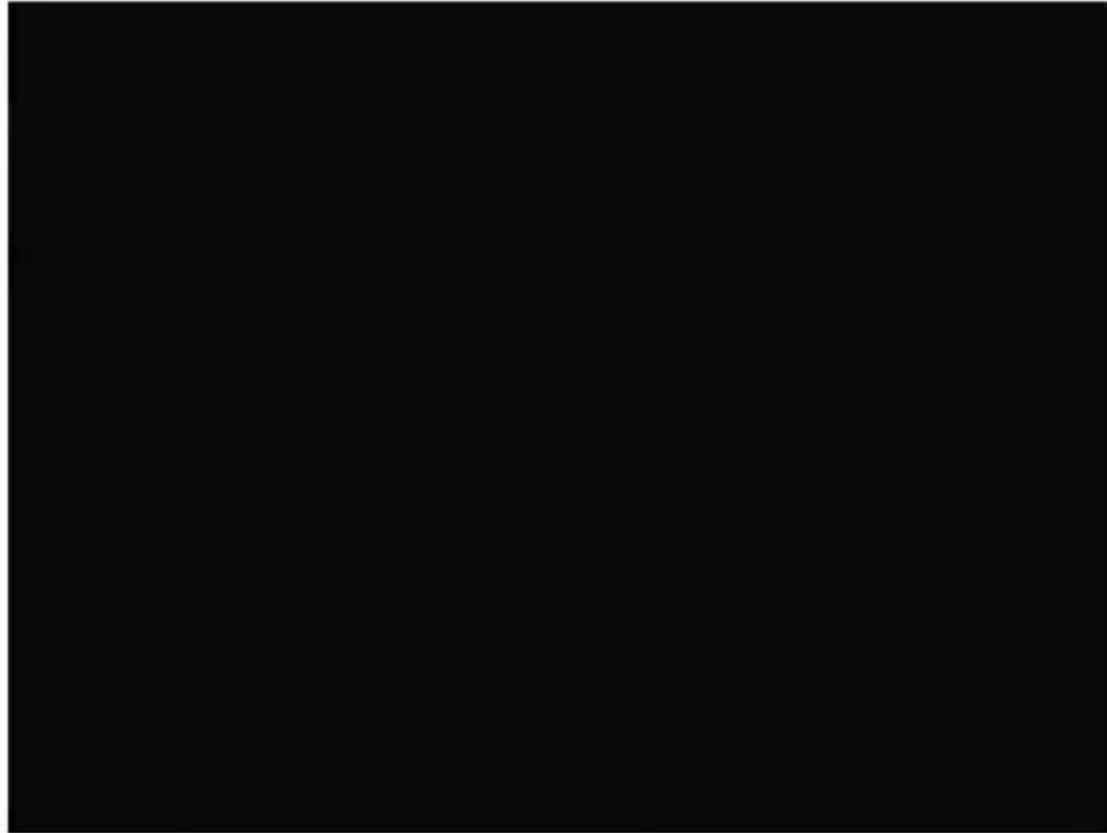


Why Don't We Already Have **Autonomous Robots?**



Berger and Wyrobeck, Stanford 2007

Why Don't We Already Have **Autonomous Robots?**

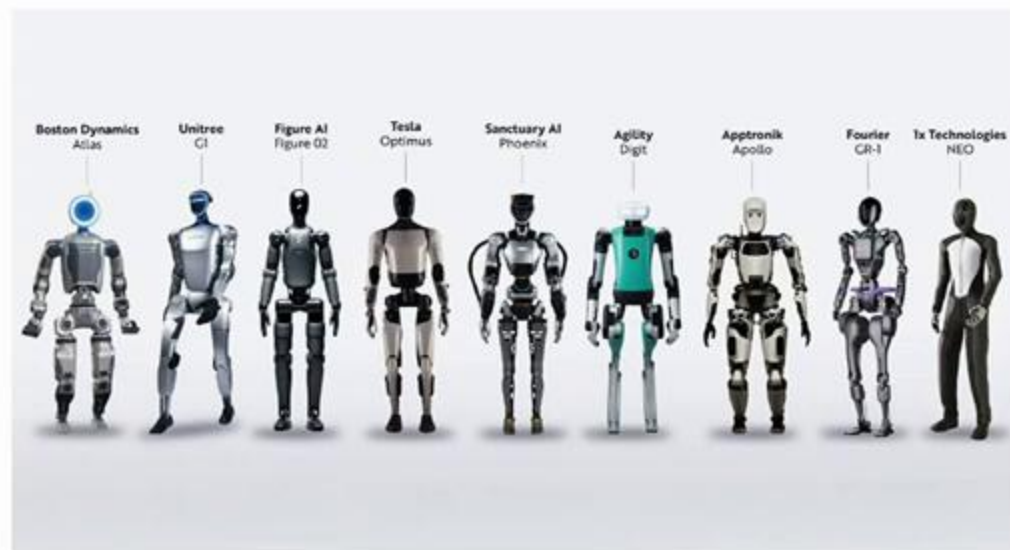


Why Don't We Already Have **Autonomous Robots?**



Robot is being
teleoperated!

Recent Hardware Advances



Humanoid Robots



NVIDIA GPUs

Recent AI Advances: **Data Scaling**

- Avg time for a human to read dataset

Imagenet



2 years

GPT-2



60 years

Llama 3

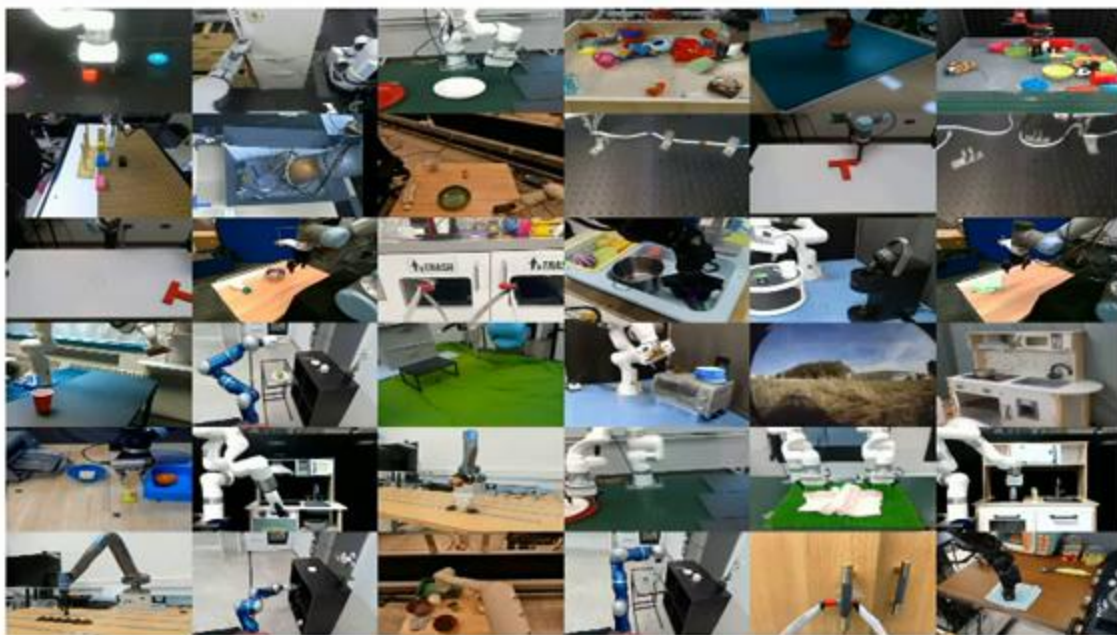
90,000 years

Recent AI Advances: **Models**



Recent Advances in Robot Learning

Large Datasets



Open X-Embodiment: Robotic Learning Datasets and RT-X Models

Open X-Embodiment Collaboration, Mees et al., ICRA, 2024.

Best Conference Paper Award (out of 1765 papers)

Large Models



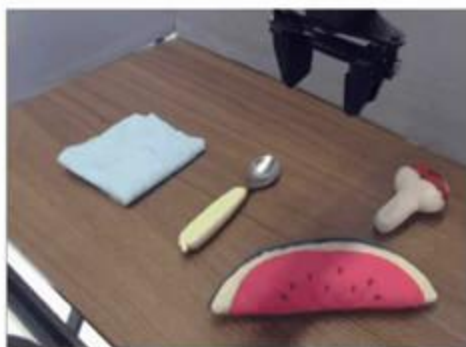
Octo

Isaac GROOT

π_0

RT-2

...



“Pick up the
spoon”

Policy



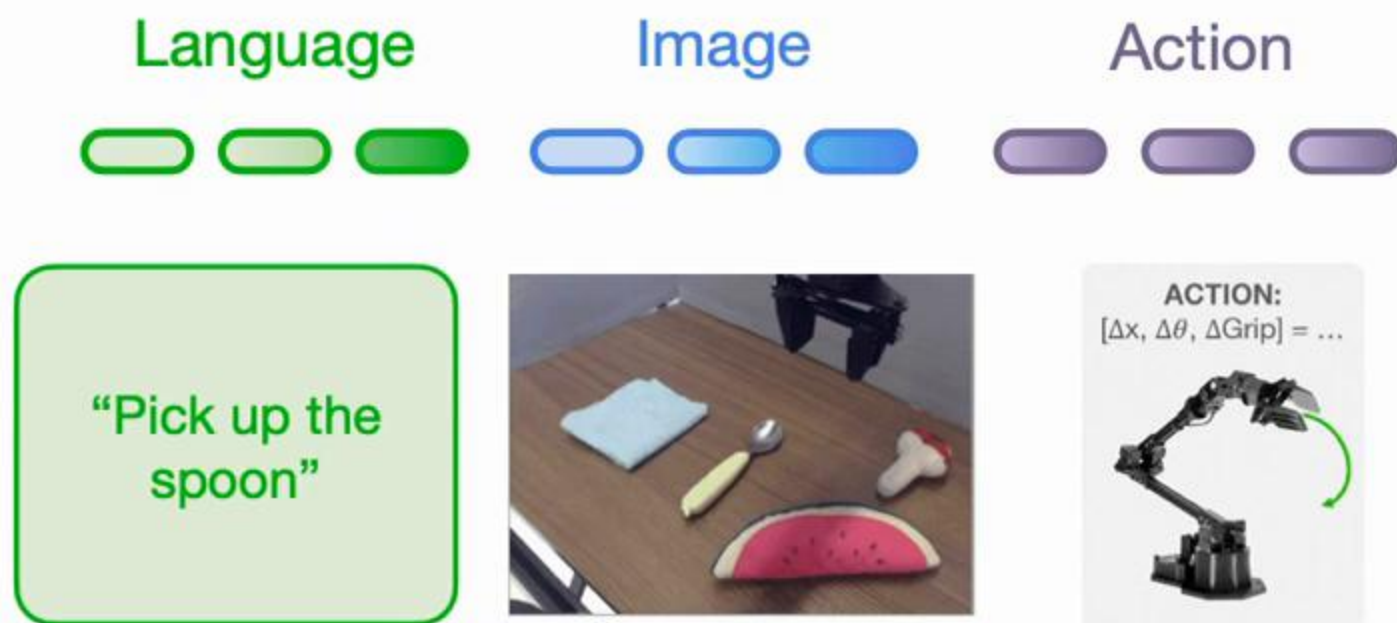


Vision-Language
Model

“Caption the
scene”

“The picture
shows
the Statue of
Liberty in NY”

Key: Robotics as Multimodal Sequence Modeling



**So we train a large transformer on robot data
and we are done?**

Ingredients for Scaling Robot Learning



Ingredients for Scaling Robot Learning

Large Datasets

Robot Data is **Scarce**

Collecting Data
requires **Human
Supervision**



Ingredients for Scaling Robot Learning

Large Datasets

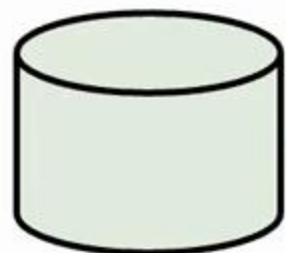
Robot Data is **Scarce**

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Large Models

Robot Data is
**Multimodal &
Heterogeneous**

Robots need **High
Frequency Control**



Ingredients for Scaling Robot Learning

Large Datasets

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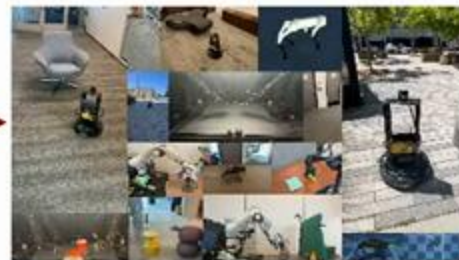
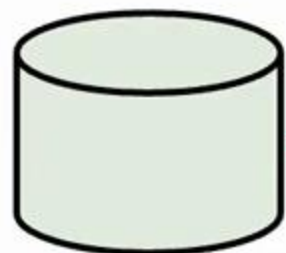
Robot Data is
**Multimodal &
Heterogeneous**

Robots need **High
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Scalable Evaluation

Robot Evals are
Tedious & Expensive

Difficult to reproduce



Algorithms for Robotic Learning

Imitation Learning

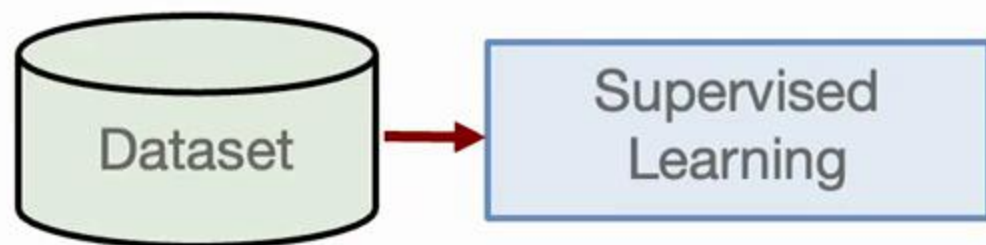
Given labeled data:

$$\mathcal{D} = \{(x_i, y_i)\},$$

learn $f(x) \approx y$

Assumes:

inputs x are independently,
identically distributed (i.i.d.)

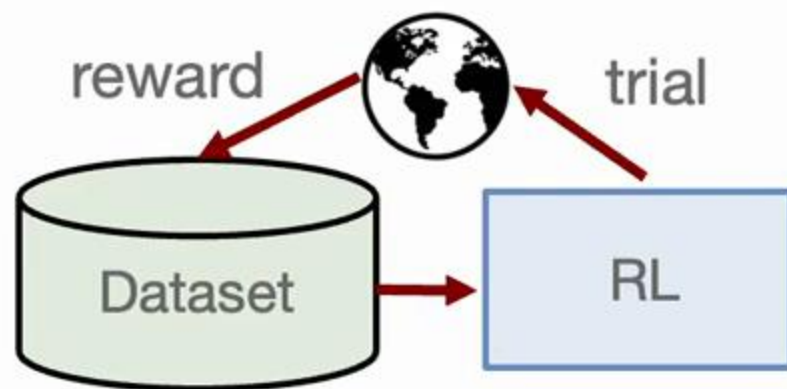


Reinforcement Learning

Learn behavior:

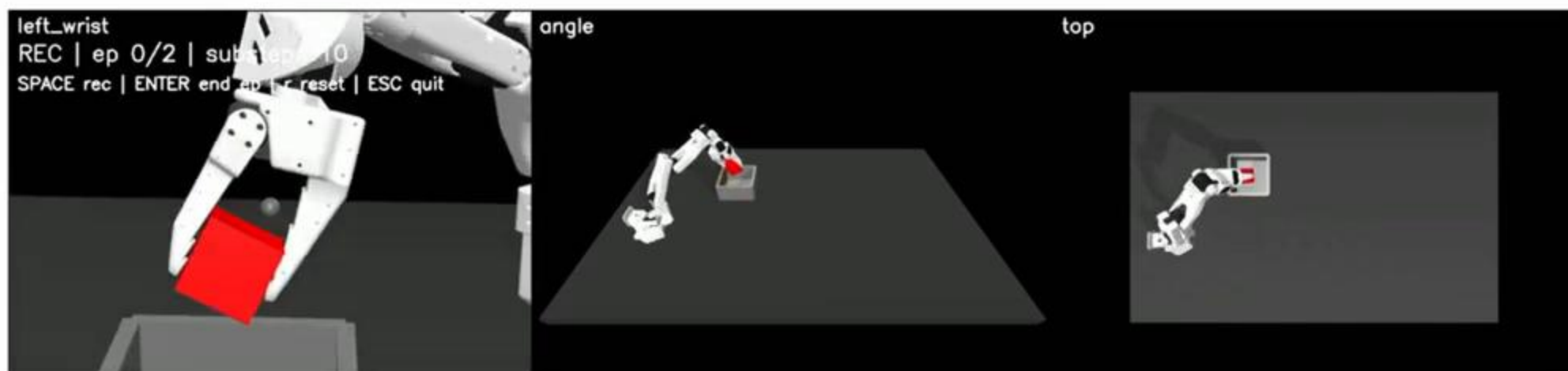
$$\pi(a|s)$$

data is **not** i.i.d., actions
affect future states



Course Syllabus: 1st Half

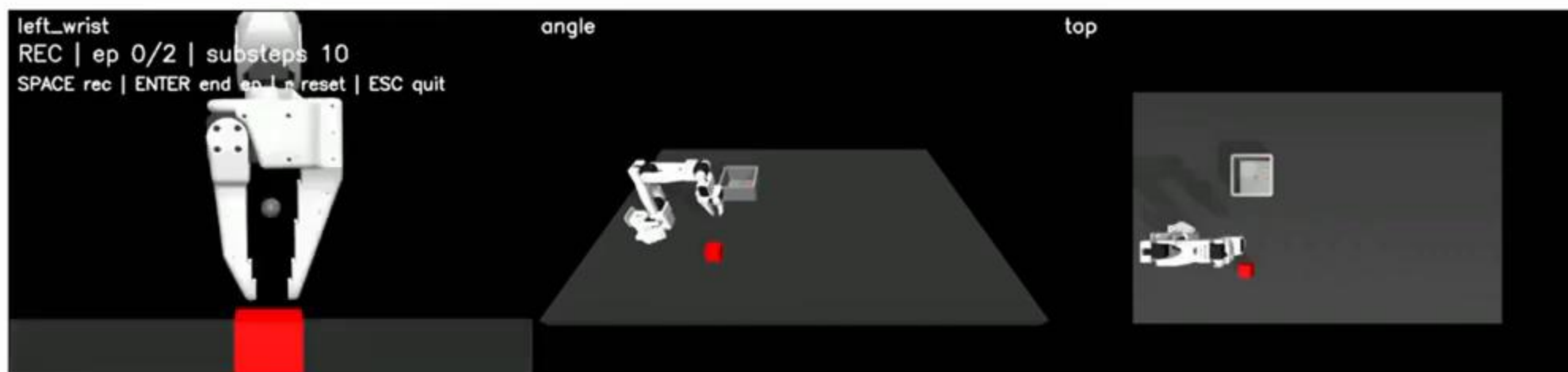
- Robot Learning Fundamentals & Algorithms
 - Robot Control & MDPs
 - Imitation Learning
 - Reinforcement Learning (Online & Offline)



Homework Assignments in Simulation

Course Syllabus: 1st Half

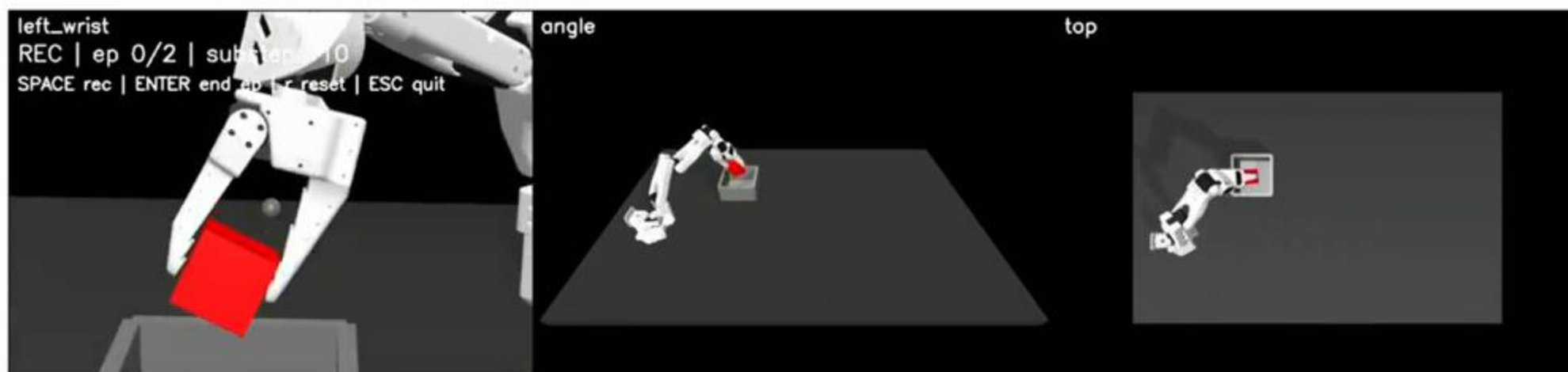
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Homework Assignments in Simulation

Course Syllabus: 1st Half

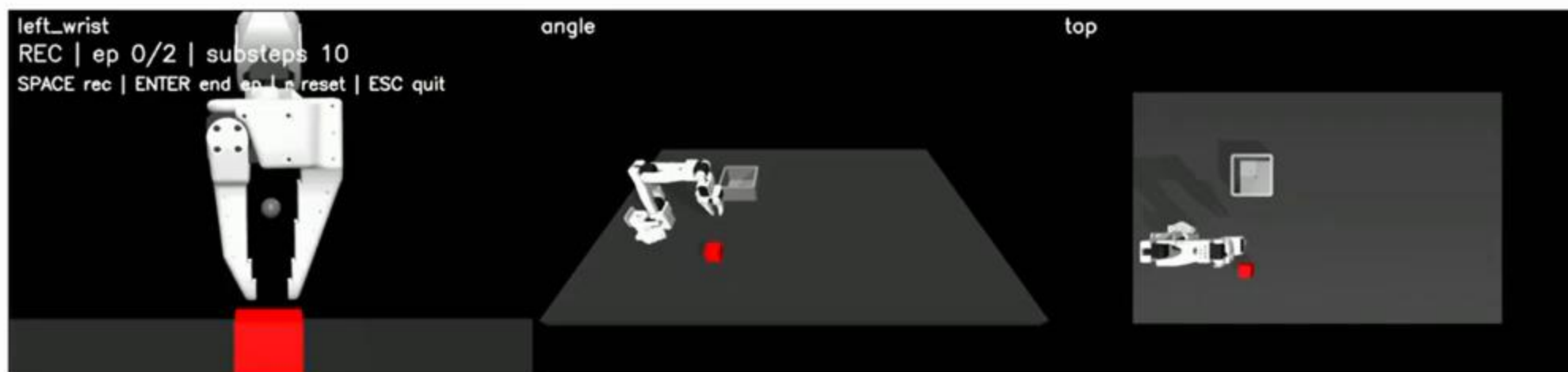
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Homework Assignments in Simulation

Course Syllabus: 1st Half

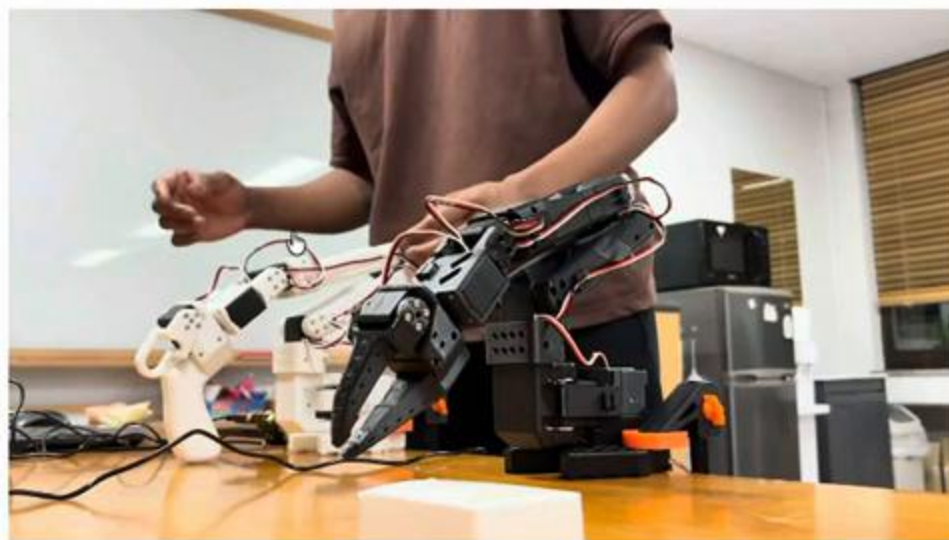
- Robot Learning Fundamentals & Algorithms
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Homework Assignments in Simulation

Course Syllabus: 2nd Half

- Scaling Robot Learning
 - Generative Models
 - Sequence Modeling & Transformers
 - World Models
 - Robot Foundation Models & Embodied Reasoning



Group Projects with Real Robots

After the Course

- You will understand:



Berger and Wyrbeck, Stanford 2007



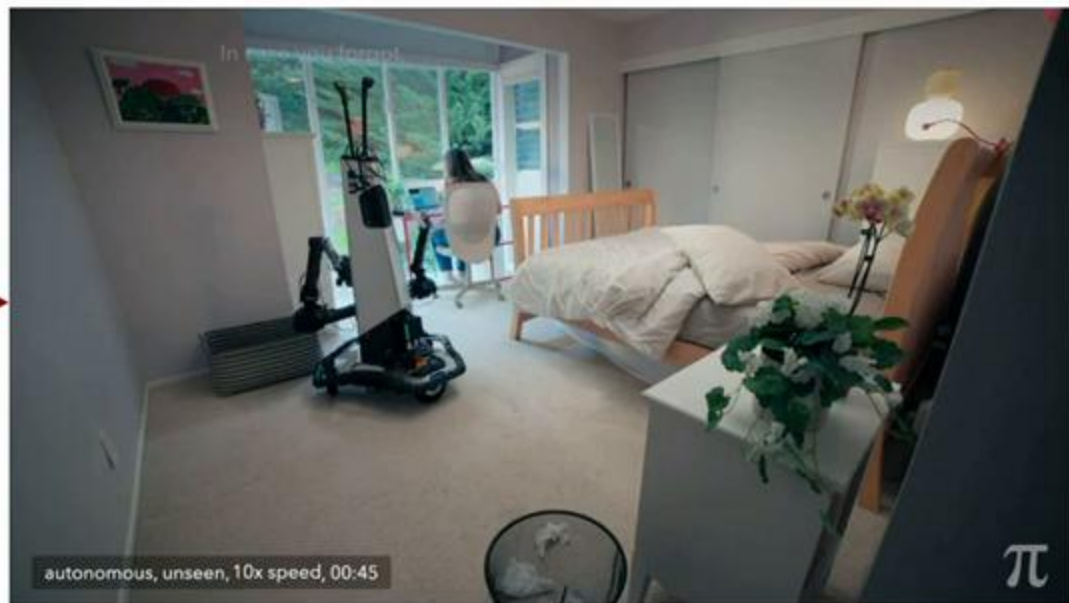
Physical Intelligence, 2025

After the Course

- You will understand:



Berger and Wyrobeck, Stanford 2007



Physical Intelligence, 2025

After the Course

- You will understand:



DARPA Robotics Challenge 2015



Figure AI, 2026

After the Course

- You will understand:



DARPA Robotics Challenge 2015



Figure AI, 2026

After the Course

- You will understand:



DARPA Robotics Challenge 2015



Figure AI, 2026

Why You Should Study Robot Learning

- Broad skills from foundation model training to PID control
- Advances often applicable to all robotic applications



Thank you for your attention
and enjoy the course!